

Ishaan Mehta

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U.S. Citizen

Education

University of Waterloo

Bachelor of Applied Science, Computer Engineering, Co-op

Waterloo, ON

September 2026 – May 2031 (expected)

Experience

AutoFillPDF

Founder & Engineer

Remote

February 2025 – Present

- Built and launched a document automation platform with \$1,000+ MRR and 70+ monthly users.
- Trained a D-FINE object detector on 13K+ annotated forms to find fields in flattened, non-fillable PDFs.
- Built the production pipeline (FastAPI, PyTorch, PostgreSQL) that pairs the detector with language models to work out what each field asks for and fill it from user context.

SoCal Rehab

ML Engineer

Remote

June 2026 – Present

- Automated clinical notes with Gemini on Vertex AI, extracting each patient's therapy data from facility PDFs and drafting follow-up notes for clinician review.
- Cut processing for a 100+ patient batch from 5 hours to 15 minutes, including billing sheets.
- Prototyped locally with Ollama, then moved to Google Apps Script so patient data stays inside the clinic's HIPAA-covered Google Workspace.

Projects

Ballform + kernelopt | Basketball shot analyzer | [GitHub](#)

September 2026

- Built a basketball computer vision system (YOLO26 pose + RF-DETR detection) that tracks players, analyzes shooting form, and measures how open each shot is.
- Ported RF-DETR to MLX with custom Metal kernels, 1.8× faster than ONNX Runtime at equal accuracy.
- Ran pose on the Neural Engine alongside GPU ball detection, cutting a 10-second clip from 180s to 25s.

MLX-VLM (5.5K+ [GitHub stars](#)) | Open-source contributor | [PRs](#)

October 2026

- Moved RF-DETR's backbone attention to MLX's fused attention kernel, making detection 21% faster.
- Fixed 4 correctness bugs, including an off-by-one that shifted detected boxes by up to 11.7 px.

Lamp Robot Simulation | [GitHub](#)

2026

- Mapped MediaPipe arm tracking onto a simulated 5-DOF URDF lamp so it follows a person's arm.

FIRST Robotics Competition

Lead Electrical Engineer & Programmer, Team 2404

Pasadena, CA

2024 – 2026

- Built a Python shoot-on-the-move targeting algorithm using vision tracking and projectile motion.
- Led electrical integration and programming for a 30+ member team.
- Won the 2026 CA District Glendale Event and placed 3rd at the Aerospace Valley Regional.

Skills

Languages: Python, TypeScript, C++, Java, SQL

ML / CV: PyTorch, MLX, Core ML, ONNX Runtime, Hugging Face, OpenCV, MediaPipe, YOLO, RF-DETR

Tools: Git, Linux, FastAPI, PostgreSQL, Google Cloud Vertex AI, Ollama, Weights & Biases, Metal